SIMATSENGINEERING
ASSESSMENTTOOL1

COURSE NAME: COMPUTER ARCHITECTURE COURSE CODE: CSA12

COURSE OUTCOME COVERED

CO1:Analyzethefunctionalunitsofacomputersystemincluding CPU,memory,bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)
AssessmentToolWeightage:40%

QUESTIONS:5 TotalMarks:25 100

ANNEXURE A
APPLICATIONBASEDQUESTIONS

12/1. 18

Code of Conduct:

I **G.THARAKNATHREDDY(192472032)** *(Name / Reg No) certify that this submission is my original work andthatIhaveadheredtotheguidelinesspecifiedforthisassessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

G. Thara

Signature of the student:

ANNEXURE A

1. A smartcitysurveillancesystemprocessescontinuousvideofeedsfrommultiplecameras,where large volumes of data are transferred between I/O devices, memory, and CPU. During peak hours, the system experiences lag and delayed response in threat detection. Analyze how the interaction between CPU, memory, and I/O contributes to this issue, and explain how different bus structures (single-bus vs multi-bus) and improvements in instruction execution cycle can enhance system performance.

A real-timestocktradingapplicationrunningonaprocessorexecutesmillionsofinstructionsper second, but usersreportdelaysintransactionprocessingduringhighload.Thesystemhasahigh CPIduetofrequentmemoryaccessandbranchinstructions.Examinehowthe fetch–decode–execute cycle impacts execution time in this scenario, and propose methods to reduce CPI and improve overall CPU performance using suitable architectural enhancements.

An embedded system based on an 8085 microprocessor is used in an industrial temperature control unit, where sensors provide input and actuatorsrespondaccordingly.However,incorrect temperature readings are occasionally processed due toimproperuseofaddressingmodesinthe program.Discusshowdifferentaddressingmodesinfluencedataaccess,identifypossiblecauses

of such errors, and suggest how the instruction set of 8085 can be effectively used to ensure accurate and reliable operation.

4. A computing system uses an 8086 microprocessor with segmented memory for executing multiple programs simultaneously, but it encounters issues like incorrect data access and stack overflow errors. Evaluate how segmentation (code, data, and stack segments) works in 8086, analyze how improper segment handling canleadtosuchproblems,andexplainhowinstruction execution and addressing modes must be managed to ensure correct program execution.

5. A data communication system represents packet information in hexadecimal format for ease of debugging and transmission, but a software engineer mistakenly interprets values, leading to incorrect data processing. Analyze the importance of number system conversions between hexadecimal and binary in such systems, explain how errorsinconversioncanaffectinstruction execution, and discuss how efficient CPU design and benchmarking can help detect and minimize such issues in real-time systems.

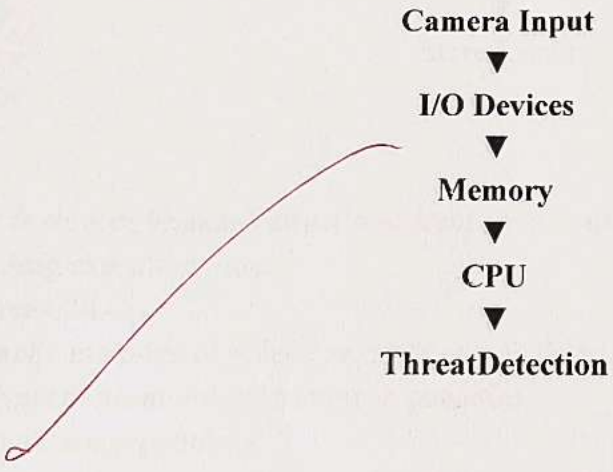
Q1.SmartCitySurveillanceSystem

Answer:

A smart city surveillance system continuously collects video data from multiple cameras. The data transferred between **I/O devices, memory, and CPU** for processing. During peak hours, thousands of vi frames are processed simultaneously, increasing system load and causing delays.

InteractionofCPU,Memory,andI/O

- **I/Odevices** (cameras) capture video and send it to the computer.
- **Memory (RAM)** temporarily stores the video frames.
- **CPU**fetchesinstructionsanddatafrommemory,processesthem,andidentifiesthreats.
- Ifthebusisbusyormemoryaccessisslow,theCPUremainsidle,reducingoverallperformance.
- **Flow chart:**



Bus Structures

Single Bus Architecture

- OnebusissharedbyCPU,memory,andI/Odevices.
- Only one data transfer occurs at a time.
- Causes congestion and increases waiting time.

Multi-Bus Architecture

- UsesseparatebusesforCPU,memory,andI/O communication.
- Multiple transfers can occur simultaneously.

- Reduces bottlenecks and improves system speed.

mprovingInstruction Execution

- Use**instruction pipelining** to execute multiple instructions simultaneously.
- Increase **cache memory** to reduce RAM access time.
- Implement**DMA(Direct MemoryAccess)**for fasterI/O transfer.
- Use faster processors and optimize scheduling.

Conclusion

Amulti-bussystem,pipelining, cache memory, and DMA significantly improve surveillance performance,reduce lag, and enable faster real-time threat detection.

Q2.Real-TimeStockTradingApplication

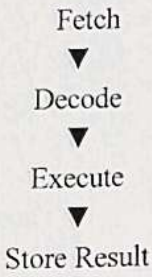
Answer:

A real-time stock trading application executes millions of instructions every second. During heavy trading, users experience delays because the processor has a **high CPI (Cycles Per Instruction)** due to frequent memory access and branch instructions.

Fetch–Decode–Execute Cycle

1. **Fetch:** CPU fetches the instruction from memory.
2. **Decode:** CPU interprets the instruction.
3. **Execute:** CPU performs the required operation.
4. **Store:**Theresult iswritten backto memoryor registers.

Flow chart:



If memory access is slow or branch instructions occur frequently, the CPU spendsmoreclockcyclesoneach instruction, increasing execution time.

Methods to Reduce CPI

- Increase **cache memory** to reduce memory access delay.
- Use**branchprediction**tominimizebranch penalties.
- Apply **instruction pipelining**.
- Use**superscalararchitecture**toexecutemultipleinstructionstogether.
- Optimize software to reduce unnecessary instructions.
- Increase processor clock frequency where possible.

Conclusion

Reducing CPI improves CPU efficiency, decreases execution time, and ensures faster transaction processing during high workloads.

Q3.8085EmbeddedTemperatureControlUnit Answer:

The **8085 microprocessor** controls industrial temperaturesystemsbyreadingsensordataandsendingcontrol signals to actuators. Correct addressing modes are essential for accurate data access.

AddressingModes

- **ImmediateAddressing:**Data isdirectlyavailable inthe instruction.
- **RegisterAddressing:**UsesCPU registersforfaster processing.
- **DirectAddressing:**Thememoryaddressisspecifiedexplicitly.
- **RegisterIndirectAddressing:**Memoryaddressisstoredinaregisterpair.

Flow chart:

Sensor
▼
ReadData
▼
8085CPU
▼
Process
▼
Actuator

CausesofIncorrectReadings

- Wrongaddressingmodeselected.
- Incorrect memory address.
- Improper sensor calibration.
- Programming errors in data transfer.
- Noiseorhardwarefaultsaffectingsensor input.

Effective Use of 8085 Instruction Set

- UseinstructionslikeMOV,MVI,LDA,STA,LXI,IN,andOUTcorrectly.
- Validatesensorreadingsbeforeprocessing.
- Store data in the correct memory locations.
- Testtheprogramunderdifferentoperatingconditions.

Conclusion

Properuseofaddressingmodesandthe 8085 instruction set ensures accurate temperature readings, reliable control, and stable industrial operation.

Q4.8086SegmentedMemory Answer:

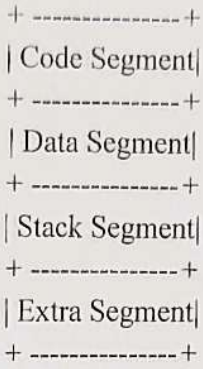
The**8086microprocessor**usessegmentedmemorytoefficientlymanageprogramsanddata.Memoryis divided into logical segments, allowing multiple programs to execute efficiently.

Memory Segments

- **Code Segment (CS):** Stores program instructions.
- **Data Segment (DS):** Stores variables and data.
- **Stack Segment (SS):** Stores return addresses and temporary data.

- **Extra Segment (ES):** Stores additional data for string operations.

Flow chart:



ProblemsDuetoImproperSegment Handling

- Incorrect segment registers access wrong memory locations.
- Stack overflow occurs when too many values are pushed without popping.
- Program crashes because of invalid memory references.
- Data corruption occurs due to overlapping segments.

ProperInstruction Execution

- Initialize all segment registers correctly.
- Use correct addressing modes.
- Balance every **PUSH** instruction with a corresponding **POP**.
- Monitor stack size during execution.
- Debug memory references carefully.

Conclusion

Proper segmentation, correct addressing, and balanced stack operations ensure reliable execution of multipl programs without memory errors.

Q5.NumberSystemConversion

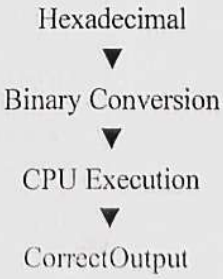
Answer:

Computers process data in **binary**, while programmers often use **hexadecimal** because it is compact a easier to understand. Accurate conversion between hexadecimal and binary is essential for debugging a communication systems.

Importance of Conversion

- Simplifies binary representation.
- Used in memory addresses and machine instructions.
- One hexadecimal digit represents **4 binary bits**.
- Makes debugging and data analysis easier.

Flow chart:



Effectsof Conversion Errors

- Wronginstructionexecution.
- Incorrect memory addressing.
- Packet transmission errors.
- Data corruption and communication failure.
- Reduced system reliability.

Role of Efficient CPU Design

- Faster CPUs execute instructions more accurately.
- Errordetection and correction mechanisms improve reliability.
- Cache memory reduces execution delays.
- BenchmarkingevaluatesCPUspeed,throughput,andefficiencyunderdifferentworkloads.

Conclusion

Correcthexadecimal-to-binaryconversion,efficientCPUarchitecture,andbenchmarkinghelpimprovesystem accuracy, detect errors early, and ensure reliable real-time communication.

MARKS ALLOCATION

Criteria	Understanding of Problem Context(20%)	Applicationof Concepts (20%)	Problem-Solving Approach(20%)	Accuracyof Solution (20%)	Clarity(20%)	
Q1	3	4	3	3	2	15
Q2	3	3	3	3	2	14
Q3	2	3	4	3	2	14
Q4	3	3	4	3	1.5	14.5
Q5	3	4	3	3	1.5	14.5

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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearlyunderstands the problem and accuratelyidentifies all requirements (4)	Goodunderstanding with minor gaps (3)	Basic understanding; some requirements unclear (2)	Poor understandingor misinterpretation of the problem (1)
Application of Concepts	30	Accuratelyapplies appropriate concepts to solve complexproblems (6)	Applies relevant conceptscorrectly withminorerrors (5)	Applies basic conceptswith limited accuracy (4)	Failstoapply appropriate concepts (1-2)

Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand